

Deliverable 1

What is a Web Server?

A **web server** is software that serves web pages to clients over the internet. It handles HTTP requests from users' browsers and delivers HTML, CSS, JavaScript, images, and other web content.

Example: Apache HTTP Server.

Different Web Server Applications

1. Apache HTTP Server

- **Definition:** Open-source web server software that is highly configurable and widely used.
- **Website/Download:** <https://httpd.apache.org/>
- **Operating System:** Windows, Linux, macOS
- **Latest Version:** 2.4.x

2. Nginx

- **Definition:** Lightweight, high-performance web server and reverse proxy.
- **Website/Download:** <https://nginx.org/>
- **Operating System:** Windows, Linux, macOS
- **Latest Version:** 1.26.x

3. Microsoft IIS (Internet Information Services)

- **Definition:** Web server software developed by Microsoft for hosting websites on Windows servers.
- **Website/Download:** <https://www.iis.net/>
- **Operating System:** Windows
- **Latest Version:** IIS 10

What is Virtualization?

Virtualization is the process of creating a virtual version of a physical device or resource, such as servers, storage, or operating systems, allowing multiple operating systems to run on a single physical machine.

What is VirtualBox?

VirtualBox is an open-source virtualization software that allows you to run virtual machines (VMs) on your computer.

- **Website/Download:** <https://www.virtualbox.org/>
- **Operating System:** Windows, Linux, macOS
- **Latest Version:** 7.1.x

What is a Virtual Machine?

A **virtual machine (VM)** is a software-based emulation of a physical computer that runs an operating system and applications as if it were a separate physical machine.

Host Machine vs Guest Machine

- **Host Machine:** The physical computer that runs virtualization software.
- **Guest Machine:** The virtual machine running inside the host.

What is Debian?

Debian is a popular open-source Linux operating system known for stability and extensive software repositories.

- **Website/Download:** <https://www.debian.org/>
- **Operating System:** Linux
- **Latest Version:** Debian 12 (Bookworm)

What is a Firewall?

A **firewall** is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

What is SSH?

SSH (Secure Shell) is a protocol used to securely connect to remote computers over a network, providing encrypted command-line access.

What is an IP Address?

An **IP address** is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

What is a Network Mask?

A **network mask (subnet mask)** defines which portion of an IP address represents the network and which part represents the host. It is used to segment networks.

What is a Port?

In networking, a **port** is a numerical identifier for specific processes or network services on a device, allowing multiple services to share the same IP address.

What is Port Forwarding?

Port forwarding is the process of redirecting communication requests from one address/port combination to another, usually to allow external access to a service inside a private network.

What is Localhost?

Localhost refers to the local computer a program is running on, usually mapped to the IP address `127.0.0.1`.

What Does 127.0.0.1 Represent?

The IP address **127.0.0.1** is the loopback address that points to the local machine, allowing a device to communicate with itself.

What is Git?

Git is a distributed version control system that tracks changes in files and allows multiple developers to collaborate on projects efficiently.

- **Website/Download:** <https://git-scm.com/>

What is GitHub?

GitHub is a web-based platform that hosts Git repositories, providing tools for collaboration, version control, and project management.

- **Website:** <https://github.com/>